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Aiden Van Greuning

Biomedical engineering graduate seeking a research-driven role with a tangible clinical endpoint. Project and lab experience across biosignal analysis, cell culture, radiation therapy simulation and experimental system design.

Skills

Biomedical Knowledge	Biomechanics, biomaterials, cardiovascular mechanics, biomedical signal processing, tissue engineering, radiation therapy physics, rehabilitation engineering, BioMEMS, medical imaging (X-ray, CT, MRI, PET)
Instrumentation & Laboratory	Cell culture (CL2), aseptic technique, biosafety protocols, microscopy, ECG/EMG/EEG acquisition and analysis, signal conditioning, sensor characterization, tensile/hardness/fatigue/impact testing, XRD, DSC
Tools & Communication	MATLAB, R, Arduino, prototyping and systems integration, technical troubleshooting, IEEE-format technical writing, literature review, engineering design documentation, oral presentation, Health Canada/FDA regulatory overview

Education

2020–2025 **Bachelor of Engineering in Biomedical Engineering**, *Toronto Metropolitan University*

Academic Projects

Capstone: Low Cost Cerebrovascular Flow Pump (Sep. 2024–Apr. 2025) Grade: 89%

- Solely executed and delivered a capstone project typically assigned to a four-person team; integrated an Arduino Due, stepper-driven peristaltic pump and flow sensor with a provided 3D-printed Circle of Willis model into a working ∼$600 steady-flow cerebrovascular simulator for interventional neuroradiology training (∼75% less cost compared to similar systems in professional labs), demonstrated at Engineering Day
- Re-scoped from pulsatile to steady flow when pump and pressure-sensor integration failed; documented the failure modes in a 40+ page report and applied Poiseuille's law to model pressure drop in the control code

ECG Signal Processing in Elderly Diabetes and Cerebromicrovascular Disease (Nov.–Dec. 2024) Grade: 75%

- Responsible for the feature-analysis stage of an electrocardiogram (ECG) classification pipeline in MATLAB: calculated heart rate and RR, ST and QT interval statistics from PQRST points in 28 clinical recordings using the Pan-Tompkins Algorithm
- The selected features fed a logistic regression model, achieving 66.7% sensitivity (20 percentage points above literature benchmarks) and 62.3% 5-fold validation accuracy, small-sample limited

Experience

Apr.–Apr. 2025 **Deputy Returning Officer**, *Elections Canada*, Richmond Hill, ON

- Verified voter ID, issued ballots and maintained accurate records for 103 electors during the 2025 federal general election; assisted with ballot counting and compliance documentation after completing Elections Canada training

Aug. 2024–Mar. 2025 **Vice President of Administration (Volunteer)**, *IEEE EMBS TMU Chapter*

- Co-led task delegation for a 7-member team delivering a case competition (co-hosted with another TMU student group) and networking events with TAs and faculty, avg. ∼70 guests per event; confirmed logistics and handled food purchasing

Sep.–Dec. 2023 **Innovation Boost Zone Apprentice**, *PINION (Innovation Boost Zone Startup, TMU)*

- Developed a volunteer-recruitment survey and researched outreach approaches for an early-stage startup supporting international student onboarding; proposed and built its first website at no cost

Certifications

2021 **Biosafety Awareness**, *Toronto Metropolitan University*

2025 **GCP–Basic; R for Data Science (Analysis & Visualization); TCPS 2: CORE (2022)**, *N2/CITI Program; LinkedIn Learning; Government of Canada*

2026 **Medical Devices; Health Canada Division 5; Lean Six Sigma Foundations**, *N2 Canada; N2/CITI Program; LinkedIn Learning*